



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Complex Variables & Laplace Transform (MAT215)

Laplace Transform Using Definition

Proof of Formulae and Piecewise Functions

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CONDUCTED BY

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Motivation of Laplace Transform

Formal Definition of Laplace Transform

Laplace Transform

The **Laplace transform** of a function $f(t)$, denoted by $\mathcal{L}\{f(t)\}$, is defined as

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

where s is a complex number and the integral converges.

Laplace Transform of Common Functions

Laplace Transform of Algebraic Functions

- $\mathcal{L}\{1\} = \frac{1}{s}$, for $\text{Re}(s) > 0$.
- $\mathcal{L}\{t\} = \frac{1}{s^2}$, for $\text{Re}(s) > 0$.
- $\mathcal{L}\{t^2\} = \frac{2}{s^3}$, for $\text{Re}(s) > 0$.
- $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, for $n \in \mathbb{N}$ and $\text{Re}(s) > 0$.
- $\mathcal{L}\{t^\alpha\} = \frac{\Gamma(\alpha + 1)}{s^{\alpha+1}}$, for $\alpha > -1$ and $\text{Re}(s) > 0$.

Problem

Prove that $\mathcal{L}\{1\} = \frac{1}{s}$, for $\text{Re}(s) > 0$, using the definition of Laplace transform.

🔍 Problem

Prove that $\mathcal{L}\{t\} = \frac{1}{s^2}$, for $\text{Re}(s) > 0$, using the definition of Laplace transform.

🔍 Problem

Prove that $\mathcal{L}\{t^2\} = \frac{2}{s^3}$, for $\text{Re}(s) > 0$, using the definition of Laplace transform.

Laplace Transform of Exponential Functions

$$\mathcal{L}\{e^{at}\} = \frac{1}{s - a}, \text{ for } \operatorname{Re}(s) > \operatorname{Re}(a).$$



Problem

Prove that $\mathcal{L}\{e^{at}\} = \frac{1}{s - a}$, for $\operatorname{Re}(s) > \operatorname{Re}(a)$, using the definition of Laplace transform.



Laplace Transform of Trigonometric Functions

- $\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$, for $\operatorname{Re}(s) > 0$.
- $\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$, for $\operatorname{Re}(s) > 0$.



Problem

Prove that $\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$, for $\operatorname{Re}(s) > 0$, using the definition of Laplace transform.



Problem

Prove that $\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$, for $\operatorname{Re}(s) > 0$, using the definition of Laplace transform.

Laplace Transform of Hyperbolic Functions

- $\mathcal{L}\{\sinh(at)\} = \frac{a}{s^2 - a^2}$, for $\operatorname{Re}(s) > |a|$.
- $\mathcal{L}\{\cosh(at)\} = \frac{s}{s^2 - a^2}$, for $\operatorname{Re}(s) > |a|$.



Problem

Prove that $\mathcal{L}\{\sinh(at)\} = \frac{a}{s^2 - a^2}$, for $\operatorname{Re}(s) > |a|$, using the definition of Laplace transform.



Problem

Prove that $\mathcal{L}\{\cosh(at)\} = \frac{s}{s^2 - a^2}$, for $\operatorname{Re}(s) > |a|$, using the definition of Laplace transform.

Laplace Transform of Piecewise Functions Using Definition

② Problem

Using the definition, find the Laplace transform of the function

$$f(t) = \begin{cases} 0, & 0 \leq t < 2, \\ 3, & 2 \leq t < 4, \\ e^t, & t \geq 4. \end{cases}$$

Problem

Using the definition, find the Laplace transform of the function



$$f(t) = \begin{cases} 0, & 0 < t < \frac{\pi}{2}, \\ 2 \cos t, & t \geq \frac{\pi}{2}. \end{cases}$$

Problem

Using the definition, find the Laplace transform of the function 

$$f(t) = \begin{cases} \cos 3t, & 0 < t < \pi, \\ 0, & t \geq \pi. \end{cases}$$

Problem

Using the definition, find the Laplace transform of the function



$$f(t) = \begin{cases} 0, & 0 \leq t < \pi, \\ \sin t, & \pi \leq t < 2\pi, \\ 0, & t \geq 2\pi. \end{cases}$$

Problem

Using the definition, find the Laplace transform of the function



$$f(t) = \begin{cases} 5 \sin t, & 0 \leq t < \pi, \\ -4 \cos(2t), & t \geq \pi. \end{cases}$$

Thank You!

We'd love your questions and feedback.

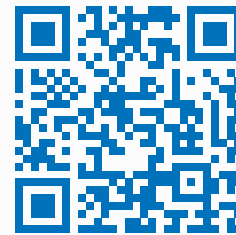
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(Lectures, walkthroughs, and course updates)



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References

- [1] Murray R. Spiegel, *Schaum's Outline of Complex Variables*, 2nd Edition, McGraw-Hill, 2009.
- [2] Dennis G. Zill, *A First Course in Differential Equations with Modeling Applications*, 11th Edition, Cengage, 2018.